



J. Dairy Sci. TBC

<https://doi.org/10.3168/jds.2025-27792>

© TBC, The Authors. Published by Elsevier Inc. on behalf of the American Dairy Science Association®.
This is an open access article under the CC BY license (<https://creativecommons.org/licenses/by/4.0/>).

Effect of adding magnesium salts or different emulsifiers to a fatty acid prill on fatty acid digestibility and milk fat production in lactating dairy cows

S. L. Burnnett,^{1*} Y. Adeniji,¹ J. Albrecht,² O. R. Drehmel,² P. Yoder,² M. F. Scott,² and K. J. Harvatine^{1†}

¹Department of Animal Science, Pennsylvania State University, University Park, PA, 16802

²Actus Nutrition, Eden Prairie, MN, 55344

ABSTRACT

Fatty acid (FA) supplements are commonly fed to increase the energy content of lactating dairy cow diets and improve milk and milk fat yield and energy balance. However, increasing dietary fat commonly decreases FA digestibility. Our objective was to investigate the effects of magnesium salts of fatty acids or exogenous emulsifiers in a FA prill on milk yield and composition and FA digestibility. Fifteen multiparous Holstein cows were randomly assigned to treatment sequence in a 5 × 5 Latin square design with 14-d periods. The treatments were (1) conventional FA prill (CON; 51.2% 16:0, 31.7% 18:0, 7.11% *cis*-9 18:1) or FA prills that contained (2) magnesium salts of FA (2.2% Mg; MAG), (3) lysophospholipids (LYSO), (4) 18:0 monoglyceride (MONO), or (5) polysorbate 60 (POLY). All cows received the same basal diet and the FA prills were fed at 2% of diet DM. Treatments had no effect on DMI, milk yield, or milk fat and protein concentration or yield. The addition of lysophospholipids increased the concentrations of preformed FA in milk compared with CON, primarily driven by an increase in *cis*-9 18:1. The addition of polysorbate increased 16 C FA digestibility compared with MAG and MONO. There was no effect of treatment on total or 18 C FA digestibility. Both MONO and POLY decreased NDF digestibility compared with CON and MAG, but there was no effect on DM digestibility. Overall, addition of emulsifiers resulted in changes in 16 C FA digestibility and preformed FA, indicating modification of FA absorption, but these did not translate to changes in milk production. Future work should explore rumen protection and possibly other types of these emulsifiers.

Key words: lysophospholipid, polysorbate, monoglyceride

INTRODUCTION

Fat is commonly supplemented to lactating dairy cows to increase the energy density of the diet. Although dietary fat supplementation generally increases milk production (Loften et al., 2014), fatty acid (FA) digestibility is often decreased with increasing dietary fat (Boerman et al., 2015). Several potential mechanisms may explain these decreases. The fat supplements may simply be less digestible than the basal diet or digestibility may be limited by either micelle formation or FA transport into intestinal cells. A limitation in the endogenous emulsifying capacity that supports micelle formation has been proposed and is currently a leading hypothesis.

Intestinal digesta in ruminants is low in UFA and monoglycerides and relies on endogenously produced bile acids and lysolecithin for formation of micelles through reduction of lipid droplet size and dissociation of FA from feed particles (Drackley, 2000). Several compounds with emulsifying properties are available for commercial use, including polysorbates, lysophospholipids, monoglycerides, and oleic acid (*cis*-9 18:1). Oleic acid is commonly included in FA prills at varying levels, but other emulsifiers have not been well investigated.

Polysorbates (PS) are nonionic surfactants made up of a sorbitol esterified with a FA. The FA has an important role in the emulsification properties of the specific compounds and is used for classification, with polysorbate 40 enriched in 16:0, polysorbate 60 (PS60) enriched in 18:0, and polysorbate 80 (PS80) enriched in *cis*-9 18:1. Prom et al. (2022) reported that a PS enriched in *cis*-9 18:1 increased FA digestibility and milk fat yield compared with no emulsifier and PS containing only 16:0 or a mixture of 16:0 and 18:0 when abomasally infused as a bolus 4×/d. The ability of abomasal infusion of *cis*-9 18:1-based PS to increase FA digestibility has been replicated in cows (de Souza et al., 2020; dos Santos Neto et al., 2023b). Overall, there is little work investigating the effects of the non *cis*-9 18:1 containing PS compounds on cow productivity, with one study finding that a blend of 16:0 and 18:0 PS has minor effects on cow performance (Prom et

Received October 19, 2025.

Accepted January 8, 2026.

*Current address: Perdue Agribusiness, Salisbury, MD 21804.

†Corresponding author: kjh182@psu.edu

The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

al., 2022). Although PS80 has an established effect on FA digestibility in cows, PS60 is currently the only form approved for feeding to livestock.

Lysophospholipids (LPPL) emulsifiers are composed of glycerol esterified to one FA and a phospholipid head. Lysolecithin is made endogenously in the duodenum of the ruminant through phospholipase cleavage of one FA from phospholipids, such as lecithin, that are present in duodenal contents. Lysophospholipids are used as an exogenous feed additive in nonruminants and have been reported to increase milk and milk fat production in sows (Zhao et al., 2017). In dairy cows, Rico et al. (2017) reported increased milk fat yield with lysolecithin added directly to a lower fat TMR and decreased milk fat in a higher fat TMR that contained soybean oil. Lee et al. (2019) reported increased milk fat yield when LPPL was added directly to the TMR. Most recently, Porter et al. (2024) reported that LPPL fed in diets with a saturated FA supplement increased total and 18 C FA digestibility but did not change milk yield.

Monoglycerides (MG) are commonly used as an emulsifier within the food industry and in calf milk replacers. Potential roles, specifically of short- and medium-chain FA derived MG as antimicrobials and promoters of gut health are being investigated in nonruminants (Jackman et al., 2022). Monoglycerides also have the potential to assist in emulsification of FA prills, but have not been explored in dairy cattle to our knowledge.

Commercial SFA prills commonly contain from trace to 12% *cis*-9 18:1, which also has emulsification properties (Freeman, 1969), and abomasal infusion of *cis*-9 18:1 has been shown to increase FA digestibility in cows (de Souza et al., 2019; Prom et al., 2021). However, if *cis*-9 18:1 is available in the rumen, it is rapidly bio-hydrogenated to 18:0 (Baldin et al., 2018), reducing its potential as an emulsifier in the small intestine. Calcium salts of FA are well investigated in dairy cows and slow the availability of FA in the rumen. Similarly, FA can be partially reacted with magnesium during the prilling process, increasing melting temperature through formation of magnesium salts (Tomkins and Lamb, 2014). The effect of Mg salts on rumen metabolism has not been explored well. Additionally, Shpirer et al. (2023) recently compared a high 16:0 free FA prill and calcium salt with the same FA profile and observed increased FA digestibility in the calcium salt (Shpirer et al., 2023). Little information is available about how magnesium salts of FA affect milk performance and FA digestibility.

If reduced FA digestibility on high-fat diets is related to a limitation in endogenously produced emulsifiers, the addition of exogenous emulsifiers may increase FA digestibility and improve efficiency of utilization of supplemented FA. The objective of this study was to investigate the effect of addition of magnesium salts of FA

or exogenous emulsifiers to FA prills on milk production and total-tract (TT) FA digestibility. We hypothesized that partially reacted Mg FA prill or the addition of an exogenous emulsifier would increase FA digestibility and milk fat yield.

MATERIALS AND METHODS

Animals and Experimental Design

All treatments and procedures were approved by the Penn State University Institutional Animal Care and Use Committee (PRAMS200946398), and the study was conducted from March through June of 2023. Fifteen multiparous cows (91 ± 33 DIM; 51 ± 7 kg/d milk; mean $\pm$ SD) from the Pennsylvania State University Dairy Research and Teaching Center (State College, PA) were housed in individual stalls. During a 14-d covariate period before the trial all cows were fed the conventional FA prill (CON; 51.3% 16:0, 31.7% 18:0, and 7.1% 18:1 n-9) diet. Cows were randomly assigned to a treatment sequence in a 5×5 Latin square design with 14-d treatment periods. Treatments were (1) CON and prills that contained either (2) Mg salts produced by melting the FA and mixing with magnesium oxide before prilling (MAG; product contained 2.2% Mg; Tomkins and Lamb, 2014), (3) lysophospholipids (LYSO; Lipidol, Pathway Intermediates, Shrewsbury, England; 1.25% of DM), (4) 18:0 monoglyceride (MONO; 6.6% of DM), or (5) PS 60 (POLY; 6.6% of DM) immediately before prilling. All prills were fed at 2% of diet DM (Table 1).

Feeding and Feed Sample Collection and Analyses

Feed was delivered to cows as a TMR once per day in the morning while cows were away at the parlor being milked. One base ration that contained ingredients common to all diets was mixed in a stationary mixer (Electra-Mix 1062; I. H. Rissler Manufacturing, Mohnton, PA), and the FA prill was added to the base mix in a mobile mixer (Rissler 1050 Feedcart; I. H. Rissler Manufacturing). Cows were fed at 105% of the previous day's intake. Dry matter of diet ingredients was calculated weekly (55°C forced-air oven for 48 h) and diets were adjusted accordingly.

Samples of diet ingredients were collected on d 12 to 14 of each period and composited by period. Refusal samples were collected on the last 2 d of each period and composited by cow within each period. Samples were stored at -20°C immediately after collection, thawed at room temperature to composite, and then stored again at -20°C until freeze-dried (Ultra 35-XL; Virtis CO. Inc., Gardiner, NY). Dried samples were ground through a 1-mm screen using a Wiley mill (A. H. Thomas Co.,

Table 1. Composition of the experimental diet, which differed only in the composition of the FA prill

Item	% DM
Ingredient	
Corn silage ¹	44.3
Alfalfa haylage ²	20.5
Canola meal	10.6
Ground corn	4.5
Heat-treated soybean meal ³	4.4
Roasted soybeans	4.0
Grass hay ⁴	3.4
Molasses	3.0
Mineral-vitamin mix ⁵	2.0
FA prill ⁶	2.0
Sodium bicarbonate	1.0
NPN ⁷	0.2
AA supplements ⁸	0.2
Composition	
CP	17.0
NDF	32.0
iNDF ₂₄₀ ⁹	12.4
Starch	21.9
Ash	7.3
Total FA	5.5

¹Corn silage contained 39.0% DM (as fed) and 39.5% NDF, 7.5% CP, 12.3% iNDF, 23.0% ADF, 33.8% starch, 2.27% ash, and 2.61% FA on a DM basis and 15.7% 16:0, 2.62% 18:0, 18.8% *cis*-9 18:1, 52.2% 18:2 n-6, and 4.76% 18:3 n-3 on an FA basis.

²Alfalfa haylage contained 41.7% DM (as fed) and 39.6% NDF, 19.8% CP, 24.0% iNDF, 35.1% ADF, 1.7% starch, and 8.79% ash on a DM basis.

³Heat-treated soybean meal was fed as SoyPlus (Landus Cooperative, Des Moines, IA).

⁴Grass hay contained 90.6% DM (as fed) and 62.6% NDF, 11.1% CP, 21.6% iNDF, 38.5% ADF, 2.1% starch, and 4.89% ash on a DM basis.

⁵Cargill Animal Nutrition.

⁶Composition of FA prills shown in Table 2.

⁷NPN fed as a coated urea (Optigen II, Alltech Inc., Lexington, KY; 256% CP on a DM basis).

⁸Rumen-protected AA fed as Smartamine M at 0.06% DM and Smartamine ML fed at 0.16% DM (Adisseo, Alpharetta, GA).

⁹Indigestible NDF measured as the residual NDF after a 240 h in vitro incubation.

Philadelphia, PA). Forages and a representative composite of concentrates in the base diet were analyzed for absolute DM by drying in a 105°C forced-air oven (AOAC 930.15; AOAC International, 2000); NDF on an organic matter basis using α amylase and sodium sulfite (Van Soest et al., 1991); ADF (AOAC 973.18; AOAC International, 2000); CP (AOAC 990.03; AOAC International, 2000); starch (Hall et al., 2015), and indigestible NDF (iNDF) by 240-h in vitro fermentation (Goering and Soest, 1970) by Cumberland Valley Analytical Services Inc. (Waynesboro, PA). Ash content was determined by complete combustion of organic material in the sample in a furnace at 500°C for 6 h. All samples were analyzed for FA composition and total FA concentration by GC after direct methylation using 10% methanolic HCl as described by Sukhija and Palmquist (1988), with modifi-

cations as described by Rico and Harvatine (2013) using 3 internal standards for determination of FA concentration (13:0 FA methyl esters [FAME], 17:1 FFA, and 19:0 FAME).

Fecal Sample Collection and Analysis

Fecal samples were collected at 6 time points (0000, 0400, 0800, 1200, 1600, and 2000 h) over the final 2 d of each period and represented approximately every 4 h across the day. Fecal samples were stored at 4°C during the collection period and composited by cow within period immediately after the final sampling. Composites were stored at −20°C, freeze-dried, ground, and analyzed for DM, ash, NDF, iNDF, and FA as described in the preceding section. Indigestible NDF was used as an internal flow marker for apparent TT digestibility calculations of DM and all other nutrients based on daily flow of fecal DM.

Milk Sampling and Analysis

Cows were milked 2×/d at approximately 0600 and 1730 h in a parlor and milk yield was determined using automated milk meters (DeLaval P500, DeLaval, Kansas City, MO). Milk samples were taken at both milkings on d 12 to 14. Samples were stored at 4°C in a bronopol-based preservative and submitted for lactose, fat, and protein analysis by mid-infrared spectroscopy (Dairy One, Ithaca, NY). On d 13, a second sample was collected and composited within day based on milk weight and 1 mL of whole milk was stored at −20°C. Whole milk samples were analyzed for FA profile as described by Rico and Harvatine (2013). Briefly, fat was extracted with hexane: isopropanol, transmethylated using sodium methoxide, and the resulting FAME quantified by GC with a flame ionization detector and a capillary column (SP-2560; 100 m × 0.25-mm internal diameter with a 0.2- μ m film thickness; Supelco Inc., Bellefonte, PA).

Statistical Analysis

Data were analyzed using the PROC MIXED in JMP Pro 17 (SAS, Cary, NC) with the following model:

$$Y_{ijkl} = \mu + C_i + P_j + T_k + e_{ijk},$$

Where Y_{ijk} represents the dependent variable of interest, μ is the overall mean, C_i is the random effect of cow ($i = 1-15$), P_j is the random effect of period ($j = 1-5$), T_k is the fixed effect of treatment ($k = 1-5$), and e_{ijk} is the residual error. Observations with Studentized residuals outside of ± 3.25 were considered outliers and removed from the analysis. Significance was claimed at $P \leq 0.05$.

and tendencies between 0.05 and ≤ 0.10 . A protected LSD was used for mean separation.

RESULTS AND DISCUSSION

Diet and Treatments

Cows were fed a common basal diet that averaged 32.0% NDF, 12.4% iNDF, 21.9% starch, and 17.0% CP across the trial (Table 1). The prills were based on a blend of approximately 51% 16:0, 32% 18:0, and 7% *cis*-9 18:1 to represent common commercially available FA prills. There were small differences in the FA content and profile of the FA prills ($P < 0.01$; Table 2). Specifically, the POLY had a slightly lower total FA concentration in both the prill (90.4% vs. 92.8%–94.9%) and TMR (5.40% vs. 5.46%–5.48%). Small differences in FA profile also occurred as a result of FA in the treatment emulsifiers. For example, the increase in 18:0 concentrations in the MONO and POLY prills was due to both containing 18:0. The increase in *cis*-9 18:1 in the MAG was likely due to variation in FA stocks during manufacturing of the prills.

Milk Yield and Milk Protein

The addition of an exogenous emulsifier did not alter milk yield ($P = 0.29$), which averaged from 40.3 to 41.7 kg/d among the treatments (Table 3). Treatment also had no effect on milk lactose concentration or yield. Prom

et al. (2022) observed no effect of abomasal infusion of PS60 or PS80 on milk yield and other recent experiments also observed no effect of PS80 (de Souza et al., 2020; dos Santos Neto et al., 2023a). Milk yield response to LPPL was more inconsistent, with 2 studies showing no effect (Rico et al., 2017; Porter et al., 2024) and one observing an increase (Lee et al., 2019). Magnesium salts of FA did not change milk yield in 3 previous experiments (Shepardson and Harvatine, 2019; Adeniji et al., 2023; Pierce, Bomberger, Burtnett, and Harvatine, unpublished data). The specific effect of MONO in FA prills on milk yield has not been explored to our knowledge, although some FA stocks used to make FA prills may contain MG.

Milk protein concentration and yield were not affected by treatment ($P = 0.25$ and 0.97 ; Table 3). Blends of 16:0 and 18:0 generally do not change milk protein concentration unless they increase milk yield. In specific investigations of emulsifiers, LPPL did not modify milk protein concentration in dairy cows (Lee et al., 2019; Porter et al., 2024) or in sows (Zhao et al., 2017). Abomasal infusion of PS has had no effect on milk protein (de Souza et al., 2020; dos Santos Neto et al., 2023a).

Milk Fat Concentration and Yield

None of the treatments affected milk fat concentration ($P = 0.20$) or yield ($P = 0.25$) in the current study (Table 3). Overall, milk fat concentration averaged 4.27% and the average milk fat in the NE milk market when the

Table 2. Fatty acid concentration and profile of TMR and FA prills

	Treatment ¹						
Item	CON	MAG	LYSO	MONO	POLY	SD	P-value ²
Diet FA profile, % of DM							
Total FA	5.46 ^a	5.48 ^a	5.48 ^a	5.48 ^a	5.40 ^b	0.13	<0.01
16 C ³	1.46 ^b	1.49 ^a	1.46 ^b	1.45 ^b	1.41 ^c	0.03	<0.01
18 C ³	3.80 ^b	3.79 ^b	3.81 ^{ab}	3.81 ^a	3.78 ^c	0.11	<0.01
16:0	1.44 ^b	1.48 ^a	1.44 ^b	1.44 ^b	1.40 ^c	0.03	<0.01
18:0	0.73 ^b	0.69 ^a	0.74 ^b	0.76 ^c	0.73 ^b	0.03	<0.01
<i>cis</i> -9 18:1	0.94 ^b	0.97 ^a	0.94 ^{bc}	0.93 ^c	0.93 ^c	0.05	<0.01
FA Prill							
Total FA, % DM	92.8 ^b	94.9 ^a	94.6 ^a	94.1 ^{ab}	90.4 ^c	2.04	<0.01
FA profile, % of FA							
16:0	51.2 ^b	52.6 ^a	50.9 ^b	50.8 ^b	50.8 ^b	0.74	<0.01
18:0	31.7 ^b	29.1 ^c	31.7 ^b	33.1 ^a	32.6 ^a	1.50	<0.01
<i>cis</i> -9 18:1	7.11 ^b	8.66 ^a	6.94 ^{bc}	6.51 ^c	6.75 ^{bc}	0.84	<0.01

^{a-c}Within XXX means that do not share a superscript differ ($P < 0.05$).

¹Treatments were a saturated FA prill that contained 51% 16:0, 32% 18:0, and 7% *cis*-9 18:1 (CON), or the same fat partially reacted with magnesium oxide (MAG) or blended with lysophospholipids (LYSO; 1.25%), 18:0 monoglyceride (MONO; 6.6%), or polysorbate 60 (POLY; 6.6%) on a DM basis before prilling.

²Effect of treatments. Means separated using a protected LSD.

³Sum of 16 carbon (16 C) and 18 carbon (18 C) FA.

Table 3. Milk production and composition of cows fed FA prills reacted with magnesium or blended with exogenous emulsifiers

Item	Treatment ¹					SEM	P-value ²
	CON	MAG	LYSO	MONO	POLY		
Milk yield, kg/d							
Milk	41.5	40.4	40.3	40.5	41.7	4.06	0.29
Fat	1.78	1.70	1.76	1.68	1.74	0.17	0.25
Protein	1.30	1.26	1.26	1.26	1.31	0.12	0.25
Lactose	1.99	1.93	1.93	1.93	2.00	0.20	0.48
Milk composition, %							
Fat	4.31	4.24	4.40	4.17	4.21	0.11	0.20
Protein	3.14	3.14	3.13	3.12	3.15	0.07	0.97
Lactose	4.79	4.77	4.80	4.78	4.78	0.04	0.94

¹Treatments were a saturated FA prill that contained 51% 16:0, 32% 18:0, and 7% *cis*-9 18:1 (CON), or the same fat partially reacted with magnesium oxide (MAG) or blended with lysophospholipids (LYSO; 1.25%), 18:0 mono-glyceride (MONO; 6.6%), or polysorbate 60 (POLY; 6.6%) on a DM basis, before prilling.

²Effect of treatments. Means separated using a protected LSD and means that do not share a superscript differ ($P < 0.05$).

study was conducted was 4.09%. This high level of milk fat in the control and treatments may indicate that dietary fat level or digestibility of FA was not limiting milk fat yield. Because all treatments contained equal amounts of dietary fat with a similar FA profile, a change in milk fat would only be expected with an increase in FA digestibility.

The effect of *cis*-9 18:1 on milk fat yield varies in the literature. Prom et al. (2021) reported that abomasal infusion of up to 60 g/d of *cis*-9 18:1 increased milk fat yield through an increase in milk yield with no change in milk fat concentration. Others observed that addition of an emulsifier to 45 g of *cis*-9 18:1 increased milk fat yield, likely a result of increases in digestibility (dos Santos Neto et al., 2023b). However, Drackley et al. (2007) reported that abomasal infusion of up to 1,000 g of high *cis*-9 18:1 sunflower oil decreased milk fat yield through a large decrease in milk yield, but Enjalbert et al. (2000) observed no effect of abomasal infusion of 400 g/d of *cis*-9 18:1. When replacing 18:0 with *cis*-9 18:1 in a blend of free FA supplements and calcium salts, FA digestibility increased but milk fat yield did not and extra energy was partitioned to body stores (de Souza et al., 2021). The current experiment was focused on FA digestibility and the short periods precluded observation of changes in body gain.

The response of milk fat to postruminal PS infusions has generally been positive, particularly when cows were provided with PS80 (de Souza et al., 2020; Prom et al., 2022), although PS80 did not increase milk yield in a recent study (dos Santos Neto et al., 2023a). When directly compared, abomasal infusion of PS80 and *cis*-9 18:1 similarly increased milk fat yield (dos Santos Neto et al., 2023b). Polysorbate 60 is esterified with 18:0, and

PS80 is esterified with *cis*-9 18:1. The current study used PS60 because it is currently approved for feed use. Prom et al. (2022) reported that abomasal infusion of 30 g of a PS containing both 16:0 and 18:0 did not increase milk fat yield, potentially because of the FA profile of the compound.

A recent study investigating LPPL in diets with a 16:0 or 18:0 based fat supplement did not observe any changes to milk fat production (Porter et al., 2024). The authors hypothesized that the basal diet was likely not energy-limiting and the additional energy from increased FA digestibility may have been partitioned to BW. Responses to LPPL have varied in other work. Rico et al. (2017) observed an increase in milk fat concentration and yield with lysolecithin added to a higher forage and lower fat diet similar to that fed in the current study. Additionally, feeding LPPL separately versus blended with a FA prill may modify its rumen metabolism and capacity in the duodenum.

Milk FA Profile

Treatments had only minor effects on milk FA profile (Table 4 and Table 5). The concentration of <16 C de novo FA and 16 C mixed-source FA was not modified by treatment ($P = 0.22$ and 0.23 , respectively). However, >16 C preformed FA were increased in the LYSO treatment compared with the MONO treatment (40.0% vs. 37.6%; $P = 0.04$) but did not differ among the other treatments. This was predominantly the result of an increase in *cis*-9 18:1, which was higher in milk fat in LYSO compared with MONO (19.0% vs. 17.5%, $P = 0.02$, Table 5) but was not different among the other treatments. These diets did not differ in the amount of *cis*-9 18:1 provided by the

Table 4. Milk FA yield and profile by source of cows fed FA prills reacted with magnesium or blended with exogenous emulsifiers

Item	Treatment ¹					SEM	P-value ²
	CON	MAG	LYSO	MONO	POLY		
FA (%)							
OBCFA ³	3.15	3.07	3.11	3.12	3.07	0.05	0.65
<16 C ⁴	24.5	24.5	23.9	25.0	24.5	0.54	0.22
16 C ⁵	30.5	30.2	29.7	30.8	30.3	0.43	0.23
>16 C ⁶	38.4 ^{ab}	37.9 ^{ab}	40.0 ^a	37.6 ^b	38.6 ^{ab}	0.78	0.04
FA (g/d)							
OBCFA	47.0	48.8	46.9	47.7	47.6	0.82	0.38
<16 C	374	370	376	374	375	10.0	0.98
16 C	460	464	470	469	456	6.69	0.37
>16 C	592	594	579	586	601	12.9	0.63

^{a,b}Within a row means that do not share a superscript differ ($P < 0.05$).

¹Treatments were a saturated FA prill that contained 51% 16:0, 32% 18:0, and 7% *cis*-9 18:1 (CON), or the same fat partially reacted with magnesium oxide (MAG) or blended with lysophospholipids (LYSO; 1.25%), 18:0 mono-glyceride (MONO; 6.6%), or polysorbate 60 (POLY; 6.6%) on a DM basis before prilling.

²Effect of treatments. Means separated using a protected LSD.

³Summation of odd- and branched-chain FA.

⁴Fatty acids <16 C originate from de novo synthesis in the mammary gland.

⁵16 C FA originate from both de novo synthesis in the mammary gland and uptake of dietary preformed FA from plasma.

⁶Fatty acids >16 C originate from the uptake of FA from plasma.

prill or in the diet, thus this may indicate an increase in absorption of 18:0 and *cis*-9 18:1. The specific difference between these treatments and lack of difference between them and the others may indicate that LYSO slightly increased absorption and MONO slightly decreased absorption relative to CON, MAG, and POLY.

Treatment tended to have an effect on *trans*-9 18:1 ($P \leq 0.10$) and *trans*-10 18:1 ($P = 0.06$) in milk fat. *Trans*-10 18:1 is a marker of the alternate biohydrogenation pathway associated with milk fat depression (Bauman and Griinari, 2001) and was 0.62% of FA in LYSO, and the other treatments ranged from 0.44% to 0.48% of FA. It is important to note that LYSO also increased >16 C preformed FA by 5% while the change in *trans*-10 18:1 was a 40% increase. However, concentrations of milk *trans*-10 18:1 were still low and expected minimal effects on milk fat concentration (Matamoros et al., 2020). Previous experiments with lecithin and lysolecithin have proposed an effect on rumen fermentation. Fontoura et al. (2021) observed a decrease in milk fat concentration when feeding lecithin, and Rico et al. (2017) reported decreased milk fat when feeding a higher starch diet that contained soybean oil. Interestingly, neither study observed increases in *trans*-10 18:1, but responses observed suggest that rumen fermentation was being altered. Milk fat odd- and branched-chain FA (OBCFA) have also been associated with changes in rumen fermentation, VFA profile, microbial populations, and milk fat depression (Fievez et al., 2012). Total OBCFA was not modified by the treatments ($P = 0.65$), although 17:0 was

slightly decreased in LYSO. The rumen availability of the LYSO when mixed in a FA prill is not known, but it appears that it may be released and has the potential to modify the rumen. Studies feeding lecithin have also reported extensive rumen biohydrogenation (Fontoura et al., 2021). Taken together, this supports the development of methods to protect the compound in the rumen beyond prilling with saturated FA.

DMI and TT DM and NDF Digestibility

Treatment did not modify DMI (Table 6, $P = 0.55$), which agrees with other investigations of PS and LPPL. Briefly, the inclusion of PS80 tended to quadratically increase intake in one study with abomasal infusion of 45 g/d (de Souza et al., 2020), but it had no effect in recent work (Prom et al., 2022; dos Santos Neto et al., 2023a). Lysophospholipids also had no effect even when fed at higher levels than the current study (Lee et al., 2019). Calcium salts of palm FA distillate consistently decreased DMI (Rabiee et al., 2012), although they contain much higher levels of unsaturated FA and other non-FA compounds than the MAG prill fed in the current study. Additionally, the MAG prill was only partially Mg salt because the reaction was limited in Mg compared with commercially available Ca salt products that are fully reacted (2% Mg vs. 10% Ca).

Treatments also had no effect on TT DM or OM digestibility ($P = 0.34$ and 0.24 , respectively), but did affect TT NDF digestibility ($P = 0.03$; Table 6). Control and

Table 5. Milk FA profile of cows fed FA prills reacted with magnesium or blended with exogenous emulsifiers

FA (%)	Treatment ¹					SEM	P-value ²
	CON	MAG	LYSO	MONO	POLY		
4:0	4.70	4.53	4.77	4.64	4.71	0.10	0.07
6:0	2.43	2.48	2.40	2.46	2.45	0.05	0.71
8:0	1.30	1.36	1.28	1.34	1.32	0.04	0.26
10:0	2.89	3.06	2.77	3.03	2.94	0.12	0.11
<i>cis</i> -9 10:1	0.24	0.24	0.23	0.26	0.23	0.01	0.13
11:0	0.05	0.05	0.04	0.05	0.05	0.01	0.14
12:0	3.20	3.40	3.04	3.41	3.25	0.16	0.07
iso 13:0	0.02	0.02	0.02	0.01	0.01	0.001	0.53
ante-iso 13:0	0.06	0.06	0.05	0.07	0.06	0.004	0.06
13:0	0.08	0.08	0.07	0.08	0.08	0.01	0.35
iso 14:0	0.06	0.07	0.06	0.07	0.06	0.01	0.59
14:0	9.98	10.10	9.57	10.16	9.88	0.24	0.08
iso 15:0	0.17	0.16	0.16	0.17	0.16	0.004	0.48
ante-iso 15:0	0.31	0.31	0.30	0.31	0.30	0.01	0.82
<i>cis</i> -9 14:1	0.74	0.72	0.72	0.82	0.71	0.04	0.19
15:0	0.81	0.80	0.79	0.81	0.80	0.03	0.94
iso 16:0	0.18	0.18	0.17	0.19	0.17	0.01	0.52
16:0	29.4	29.1	28.5	29.7	29.2	0.43	0.20
iso 17:0	0.41	0.40	0.43	0.40	0.39	0.02	0.52
16:1c9	1.10	1.10	1.15	1.14	1.13	0.04	0.84
ante-iso 17:0	0.35	0.34	0.33	0.34	0.33	0.02	0.85
17:0	0.50 ^{ab}	0.46 ^c	0.52 ^a	0.48 ^{bc}	0.50 ^{ab}	0.01	<0.01
<i>cis</i> -10 17:1	0.15	0.14	0.16	0.14	0.15	0.01	0.17
18:0	12.1	11.9	12.2	11.7	12.0	0.45	0.67
<i>trans</i> -4 18:1	0.02	0.02	0.02	0.02	0.02	0.001	0.39
<i>trans</i> -5 18:t5	0.02	0.02	0.02	0.02	0.02	0.001	0.90
<i>trans</i> -6–8 18:1	0.32	0.31	0.35	0.32	0.32	0.01	0.12
<i>trans</i> -9 18:1	0.28	0.27	0.30	0.27	0.28	0.01	>0.10
<i>trans</i> -10 18:1	0.48	0.44	0.62	0.47	0.48	0.05	0.06
<i>trans</i> -11 18:1	0.93	0.98	1.01	0.94	0.94	0.05	0.62
<i>trans</i> -12 18:1	0.48	0.41	0.51	0.53	0.53	0.04	0.14
<i>cis</i> -9 18:1	18.0 ^{ab}	17.8 ^{ab}	19.0 ^a	17.5 ^b	18.2 ^{ab}	0.48	0.02
<i>cis</i> -11 18:1	0.99	0.97	1.08	0.99	1.01	0.04	0.24
<i>cis</i> -12 18:1	0.45	0.46	0.48	0.47	0.46	0.02	0.75
18:2 n-6	2.72	2.72	2.79	2.75	2.69	0.07	0.78
20:0	0.18	0.18	0.18	0.18	0.18	0.01	0.62
18:3 n-6	0.50	0.48	0.50	0.51	0.49	0.02	0.25
<i>cis</i> -11 20:1	0.07	0.07	0.07	0.07	0.07	0.01	0.77
<i>cis</i> -9 <i>trans</i> -11 18:2	0.41	0.41	0.44	0.42	0.40	0.02	0.37
20:2 n-6	0.02	0.02	0.02	0.02	0.02	0.001	0.43
22:0	0.14	0.14	0.13	0.14	0.14	0.01	0.33
20:3 n-6	0.05	0.05	0.05	0.05	0.05	0.002	0.82
<i>cis</i> -13 22:1	0.14	0.14	0.14	0.15	0.14	0.01	0.42
20:5 n-3	0.01	0.02	0.02	0.02	0.01	0.01	0.61
24:0	0.03	0.04	0.03	0.03	0.03	0.002	0.95
20:5 n-3	0.01	0.01	0.02	0.02	0.02	0.01	0.06
22:5 n-3	0.07	0.07	0.07	0.07	0.06	0.003	0.76

^{a-c}Within rows means that do not share a superscript differ ($P < 0.05$).

¹Treatments were a saturated FA prill that contained 51% 16:0, 32% 18:0, and 7% *cis*-9 18:1 (CON), or the same fat partially reacted with magnesium oxide (MAG) or blended with lysophospholipids (LYSO; 1.25%), 18:0 mono-glyceride (MONO; 6.6%), or polysorbate 60 (POLY; 6.6%) on a DM basis before prilling.

²Effect of treatments. Means separated using a protected LSD.

MAG had the highest NDF digestibility, while MONO and POLY decreased it by more than 3 percentage points relative to CON. Data on the effect of emulsifiers in the rumen is very limited (see review by Chacko, 2020). Emulsifiers, such as PS, have several potential modes of action in the rumen, including the ability to assist in digestion by increasing enzyme activity, interaction of enzyme and substrate, and enzyme activity and

modify microbial populations. For example, Kamande et al.(2000) observed that both PS60 and PS80 increased protease and cellulase activity of rumen microbial enzymes through multiple potential mechanisms, leading to an increase in the rate and extent of cellulose degradation, with more effective results with the addition of PS80. However, the doses and possible negative effects on rumen microbes are not well explored. Abomasal in-

Table 6. Nutrient digestibility of cows fed FA prills reacted with magnesium or blended with exogenous emulsifiers

Item	Treatment ¹					SEM	P-value ²
	CON	MAG	LYSO	MONO	POLY		
Intake, kg/d							
DM	30.4	29.3	29.9	29.8	30.7	2.54	0.55
FA	1.59	1.6	1.63	1.61	1.64	0.07	0.78
16C FA	0.42	0.44	0.43	0.42	0.43	0.02	0.78
18C FA	1.11	1.12	1.14	1.12	1.15	0.05	0.67
Fecal FA							
Total, %	5.45 ^{ab}	5.31 ^{abc}	5.05 ^{bc}	5.55 ^d	4.99 ^c	0.27	0.03
16C FA, % of FA	27.1 ^b	29.2 ^a	27.0 ^b	28.5 ^a	25.0 ^c	0.54	<0.01
18C FA, % of FA	64.5 ^b	62.5 ^c	64.6 ^b	63.2 ^c	66.6 ^a	0.63	<0.01
Digestibility, %							
DM	64.8	65.3	64.3	63.9	64.5	1.73	0.34
OM	65.8	66.5	65.4	64.8	65.6	1.68	0.24
NDF	43.7 ^a	44.0 ^a	43.0 ^{ab}	41.6 ^b	41.7 ^b	1.33	0.03
FA	65.4	66.5	67.2	63.4	67.3	2.26	0.08
16C FA	64.8 ^{abc}	64.2 ^{bc}	66.6 ^{ab}	60.6 ^c	68.7 ^a	2.31	<0.01
18C FA	67.9	69.7	69.4	66.8	69.0	2.13	0.24

^{a-d}Within rows means that do not share a superscript differ ($P < 0.05$).

¹Treatments were a saturated FA prill that contained 51% 16:0, 32% 18:0, and 7% *cis*-9 18:1 (CON), or the same fat partially reacted with magnesium oxide (MAG) or blended with lysophospholipids (LYSO; 1.25%), 18:0 mono-glyceride (MONO; 6.6%), or polysorbate 60 (POLY; 6.6%) before prilling.

²Effect of treatments. Means separated using a protected LSD.

fusion of PS has shown no change with PS80 (de Souza et al., 2020) or increases in NDF digestibility with PS40 and PS80 (Prom et al., 2022; dos Santos Neto et al., 2023b). Abomasal infusion eliminated possible negative effects in the rumen but can still modify rumen motility and retention time. It is also essential to note that a difference in the type of PS may also be important. The present study indicated that blending PS60 with a saturated FA may result in reduced fiber digestion, providing additional support for rumen protection. The decrease in NDF digestion was not large enough to change DMI and does not appear to have affected acetate supply, as indicated by no change in milk de novo and 16 C FA. Future work is also needed to better understand the potential for positive and negative effects on rumen fermentation.

Fatty Acid Intake and TT FA Digestibility

Despite the small differences in the FA concentrations among the diets, as discussed previously, there were no differences in total FA intake or intake of 16 and 18 C FA (Table 6). Furthermore, addition of magnesium salts of FA or an emulsifier to the FA prill had no effect on total FA or 18 C FA digestibility ($P = 0.08$ and 0.48 , respectively), but there was an effect of treatment on 16 C FA digestibility ($P < 0.01$; Table 6). Total-tract 16 C FA digestibility was highest with the POLY treatment, and was increased by 8.1 percentage units compared with MONO and 4.5 percentage units compared with MAG. Although

16 C digestibility was altered with treatments, this did not translate to differences in milk production. We expect the increase in FA intake had implications for physiological processes outside of the scope of this experiment. For example, the current study was not designed to measure effects on BW, BCS, reproduction, or immune function; however, during long-term feeding and in a commercial setting these may be affected and have important effects on production and profitability.

Polysorbate 80 has been reported to increase total, 16 C, and 18 C FA digestibility when abomasally infused (de Souza et al., 2020; dos Santos Neto et al., 2023b). Ruminant infusion of PS80 similarly increased FA digestibility (dos Santos Neto et al., 2023a). The rumen availability and rumen bypass value of PS blended in a FA prill is not clear but did result in increased FA digestibility in the current study. Note that previous literature has focused on the use of PS80 as an exogenous emulsifier compound because of the greater benefit observed in FA digestibility observed by Prom et al. (2022). However, there are concerns about the effect of PS80 in humans because of potential contributions to intestinal inflammation, possibly as a result of breakdown of intestinal mucus barriers (Chassaing et al., 2015). The PS80 is not expected to be transferred to milk, but feeding would create similar questions on intestinal health for the cow. In the current study, PS60 was selected because it is approved for feed use and provides an opportunity to increase FA digestibility, although the negative impact on NDF digestibility will require additional investigation.

Table 7. Fatty acid absorption and transfer of 18 C FA of cows fed FA prills reacted with magnesium or blended with exogenous emulsifiers

18 C FA	Treatment ¹					SEM	P-value ²
	CON	HM	LYSO	MONO	POLY		
Intake, kg/d	1.11	1.12	1.13	1.12	1.15	0.05	0.67
Absorbed, kg/d	0.74	0.76	0.78	0.75	0.79	0.04	0.48
Milk yield, kg/d	0.61	0.59	0.58	0.57	0.61	0.03	0.26
FA balance							
Intake - milk, kg/d	0.50	0.53	0.55	0.55	0.54	0.05	0.60
Absorbed - milk, ⁴ kg/d	0.13	0.17	0.20	0.18	0.18	0.04	0.37
FA transfer to milk							
% FA consumed	55.5	52.3	52.0	51.3	53.7	3.05	0.46
% FA digested	80.0	80.9	75.9	77.1	78.5	4.60	0.78

¹Treatments were a saturated FA prill that contained 51% 16:0, 32% 18:0, and 7% *cis*-9 18:1 (CON), or the same fat reacted with magnesium oxide (MAG) or blended with lysophospholipids (LYSO; 1.25%), 18:0 monoglyceride (MONO; 6.6%), or polysorbate 60 (POLY; 6.6%) before prilling.

²Effect of treatments. Means separated using a protected LSD.

Limited research exists on MG as a feed additive, with the work primarily focused on the potential of short- and medium-chain monoglycerides as antimicrobial and antiviral additives in nonruminant animals. The FA in the monoglyceride is important to the physical properties and biological function and most of the previous research in nonruminant animals has specifically investigated MG that contain butyric (4:0), caprylic (8:0), capric (10:0), and lauric (12:0) acids, while the present study used an 18:0-based MG that is commonly used in calf milk replacers. Several studies have shown that a blend of MG (primarily composed of 4:0, 8:0, and 10:0) increased growth in broiler chickens (Feye et al., 2021) and improved immune response during a disease challenge (Daneshmand et al., 2023). Although nutrient digestibility is not often measured in these studies, gross energy digestibility increased with the supplementation of a MG blend of 4:0, 8:0, and 10:0 during a necrotic enteritis challenge (Kumar et al., 2021). Given the relationship between health and performance, it is possible that the benefits of MG are less evident in healthy animals, as shown in the lack of improvements in a study with nonchallenged broilers (Sacakli et al., 2023). It is possible that other MG would be more effective emulsifiers compared with the one used in the present study and this warrants additional research.

FA Absorption and Transfer to Milk

The 18 C FA in milk fat are assumed to originate from plasma, and the efficiency of transfer of 18 C FA from feed to milk appears to be dependent on the amount of 18 C FA consumed, with higher transfer at lower feeding rates (Khiaosa-ard et al., 2015). There were no effects of treatment on 18 C FA intake, amount absorbed, or total amount transferred to milk fat (Table 7). Additionally the

18 C FA balance based on FA intake or digestibility was unaffected ($P = 0.60$ and 0.37). Transfer of consumed or absorbed 18 C FA to milk fat was also unaffected ($P = 0.46$ and 0.78). The transfer ranged from 51.3% to 55.5% of FA consumed and is similar to the average transfer rates of 18 C FA in abomasal infusion studies (Enjalbert et al., 2000) and the plateau of transfer observed in moderate fat diets reported by (Glasser et al., 2008a). The lack of increase in milk production and milk fat with treatments that increased FA digestibility in the current experiment may be the result of this comparatively adequate FA status relative to metabolic need in all treatments.

CONCLUSIONS

There were no major effects of feeding a magnesium salt of FA or LPPL to production or digestibility measurements. More importantly, this work demonstrates the potential of exogenous emulsifiers blended in an SFA prill to affect FA digestibility. Specifically, the addition of PS60 increased 16 C FA digestibility, though it decreased NDF digestibility, indicating that rumen protection may be necessary. Blending an 18:0 MG with the prill decreased 16 C digestibility as well as NDF digestibility. The digestibility differences did not translate to production responses. The limited research with PS and MG compounds, especially in the ruminant animal, and the availability of multiple types of each compound, provide the opportunity for additional approaches that could help optimize FA digestibility.

NOTES

The work was supported by a Foundation for Food and Agriculture (FFAR, Washington DC) graduate fellowship to SLB collaboratively funded by Actus Nutrition

(previously Milk Specialties Global, Eden Prairie, MN). We also recognize support from Pennsylvania State University (University Park, PA), including USDA National Institute of Food and Agriculture (Washington, DC) federal appropriations under Project PEN04664 accession 1017181, PEN04678 accession 1017801, and PEN04907 accession 7005703. The authors extend their gratitude to the Pennsylvania State University Dairy Cattle Research and Education Center for excellent care of the animals, as well as Muhammad Arif, Abiel Haile, Muhammad Husnain, Cesar Matamoros, and Alanna Homan and to the undergraduate student workers of the Harvatin research group who helped with this trial (all from The Pennsylvania State University, University Park, PA). The FA prills were donated by Actus Nutrition (Eden Prairie, MN), the SoyPlus expeller soybean meal was donated by Landus Cooperative (Ames, IA), and the amino acid supplements were donated by Adisseo (Alpharetta, GA). All treatments and procedures were approved by the Pennsylvania State University Institutional Animal Care and Use Committee (PRAMS200946398). Harvatine is a member of Actus Nutrition's scientific advisory board. Burtnett is currently employed by Perdue Agribusiness (Salisbury, MD), and Albrecht, Drehmel, Yoder, and Scott are employees of Actus Nutrition. The authors have not stated any other conflicts of interest.

Nonstandard abbreviations used: CON = conventional FA prill; FA = fatty acids; FAME = FA methyl esters; iNDF = indigestible NDF; LPPL = lysophospholipids; LYSO = treatment including Lipidol at 1.25% of DM; MAG = treatment including Mg salts produced by melting the FA and mixing with magnesium oxide before prilling; MG = monoglycerides; MONO = treatment including 18:0 monoglyceride at 6.6% of DM; OBCFA = odd- and branched-chain fatty acids; POLY = treatment including PS60 at 6.6% of DM; PS = polysorbate; TT = total tract.

REFERENCES

- Adeniji, Y. A., R. Bomberger, and K. Harvatine. 2023. Interaction of pretrial milk fat production and dietary fat supplementation on milk and milk fat yield in Holstein cows. *J. Dairy Sci.* 106(Suppl. 1):169. (Abstr.)
- AOAC International. 2000. Official Methods of Analysis. 17th ed. AOAC International, Gaithersburg, MD.
- Baldin, M., D. E. Rico, M. H. Green, and K. J. Harvatine. 2018. *Technical note*: An in vivo method to determine kinetics of unsaturated fatty acid biohydrogenation in the rumen. *J. Dairy Sci.* 101:4259–4267. <https://doi.org/10.3168/jds.2017-13452>.
- Bauman, D. E., and J. M. Griinari. 2001. Regulation and nutritional manipulation of milk fat: Low-fat milk syndrome. *Livest. Prod. Sci.* 70:15–29. [https://doi.org/10.1016/S0301-6226\(01\)00195-6](https://doi.org/10.1016/S0301-6226(01)00195-6).
- Boerman, J. P., J. L. Firkins, N. R. St-Pierre, and A. L. Lock. 2015. Intestinal digestibility of long-chain fatty acids in lactating dairy cows: A meta-analysis and meta-regression. *J. Dairy Sci.* 98:8889–8903. <https://doi.org/10.3168/jds.2015-9592>.
- Chassaing, B., O. Koren, J. K. Goodrich, A. C. Poole, S. Srinivasan, R. E. Ley, and A. T. Gewirtz. 2015. Dietary emulsifiers impact the mouse gut microbiota promoting colitis and metabolic syndrome. *Nature* 519:92–96. <https://doi.org/10.1038/nature14232>.
- Daneshmand, A., N. K. Sharma, S. K. Kheravii, L. Hall, and S.-B. Wu. 2023. Buffered formic acid and a monoglyceride blend improve performance and modulate gut bacteria and immunity gene expression in broilers under necrotic enteritis challenge. *Poult. Sci.* 102:102978. <https://doi.org/10.1016/j.psj.2023.102978>.
- de Souza, J., C. M. Prom, and A. L. Lock. 2021. Altering the ratio of dietary palmitic and oleic acids affects nutrient digestibility, metabolism, and energy balance during the immediate postpartum in dairy cows. *J. Dairy Sci.* 104:2910–2923. <https://doi.org/10.3168/jds.2020-19312>.
- de Souza, J., N. R. St-Pierre, and A. L. Lock. 2019. Altering the ratio of dietary C16:0 and cis-9 C18:1 interacts with production level in dairy cows: Effects on production responses and energy partitioning. *J. Dairy Sci.* 102:9842–9856. <https://doi.org/10.3168/jds.2019-16374>.
- de Souza, J., M. Western, and A. L. Lock. 2020. Abomasal infusion of an exogenous emulsifier improves fatty acid digestibility and milk fat yield of lactating dairy cows. *J. Dairy Sci.* 103:6167–6177. <https://doi.org/10.3168/jds.2020-18239>.
- dos Santos Neto, J. M., C. M. Prom, and A. L. Lock. 2023a. Comparison of rumen and abomasal infusions of an exogenous emulsifier on fatty acid digestibility of lactating dairy cows. *J. Dairy Sci.* 106:6789–6797. <https://doi.org/10.3168/jds.2022-23143>.
- dos Santos Neto, J. M., C. M. Prom, and A. L. Lock. 2023b. Abomasal infusion of oleic acid and exogenous emulsifier alter fatty acid digestibility and production responses of lactating dairy cows. *J. Dairy Sci.* 106:7591–7601. <https://doi.org/10.3168/jds.2022-23038>.
- Drackley, J. K. 2000. Lipid metabolism. Pages XXX–XXX in XXX.
- Drackley, J. K., T. R. Overton, G. Ortiz-Gonzalez, A. D. Beaulieu, D. M. Barbano, J. M. Lynch, and E. G. Perkins. 2007. Responses to increasing amounts of high-oleic sunflower fatty acids infused into the abomasum of lactating dairy cows. *J. Dairy Sci.* 90:5165–5175. <https://doi.org/10.3168/jds.2007-0122>.
- Enjalbert, F., M. C. Nicot, C. Bayourthe, and R. Moncoulon. 2000. Effects of duodenal infusions of palmitic, stearic, or oleic acids on milk composition and physical properties of butter. *J. Dairy Sci.* 83:1428–1433. [https://doi.org/10.3168/jds.S0022-0302\(00\)75012-0](https://doi.org/10.3168/jds.S0022-0302(00)75012-0).
- Feye, K. M., D. K. Dittoe, J. A. Jendza, J. P. Caldas-Cueva, B. A. Mallmann, B. Booher, G. Tellez-Isaias, C. M. Owens, M. T. Kidd, and S. C. Ricke. 2021. A comparison of formic acid or monoglycerides to formaldehyde on production efficiency, nutrient absorption, and meat yield and quality of Cobb 700 broilers. *Poult. Sci.* 100:101476. <https://doi.org/10.1016/j.psj.2021.101476>.
- Fievez, V., E. Colman, J. M. Castro-Montoya, I. Stefanov, and B. Vlaeminck. 2012. Milk odd- and branched-chain fatty acids as biomarkers of rumen function—An update. *Anim. Feed Sci. Technol.* 172:51–65. <https://doi.org/10.1016/j.anifeedsci.2011.12.008>.
- Fontoura, A. B. P., J. E. Rico, A. N. Davis, W. A. Myers, B. N. Tate, R. Gervais, and J. W. McFadden. 2021. Effects of dietary deoiled soy lecithin supplementation on milk production and fatty acid digestibility in Holstein dairy cows. *J. Dairy Sci.* 104:1823–1837. <https://doi.org/10.3168/jds.2020-18797>.
- Freeman, C. P. 1969. Properties of fatty acids in dispersions of emulsified lipid and bile salt and the significance of these properties in fat absorption in the pig and the sheep. *Br. J. Nutr.* 23:249–263. <https://doi.org/10.1079/BJN19690032>.
- Glasser, F., A. Ferlay, M. Doreau, P. Schmidely, D. Sauvant, and Y. Chilliard. 2008a. Long-chain fatty acid metabolism in dairy cows: A meta-analysis of milk fatty acid yield in relation to duodenal flows and de novo synthesis. *J. Dairy Sci.* 91:2771–2785. <https://doi.org/10.3168/jds.2007-0383>.
- Glasser, F., P. Schmidely, D. Sauvant, and M. Doreau. 2008b. Digestion of fatty acids in ruminants: A meta-analysis of flows and variation factors: 2. C18 fatty acids. *Animal* 2:691–704. <https://doi.org/10.1017/S1751731108002036>.
- Goering, H. K., and P. J. V. Soest. 1970. Forage Fiber Analyses (Apparatus, Reagents, Procedures, and Some Applications). Agriculture Handbook No. 379. U.S. Agricultural Research Service. <https://doi.org/10.1017/S1751731108002036>.

- www.govinfo.gov/content/pkg/GOVPUB-A-PURL-gpo24229/pdf/GOVPUB-A-PURL-gpo24229.pdf.
- Hall, M. B., J. Arbaugh, K. Binkerd, A. Carlson, T. Doan, T. Grant, C. Heuer, H. D. Inerowicz, B. Jean-Louis, R. Johnson, J. Jordan, D. Kondratko, E. Maciel, K. McCallum, D. Meyer, C. A. Odijk, A. Parganlija-Ramic, T. Potts, L. Ruiz, S. Snodgrass, D. Taysom, S. Trupia, B. Steinlicht, and D. Welch. 2015. Determination of dietary starch in animal feeds and pet food by an enzymatic-colorimetric method: Collaborative study. *J. AOAC Int.* 98:397–409. <https://doi.org/10.5740/jaoacint.15-012>.
- Jackman, J. A., T. A. Lavergne, and C. C. Elrod. 2022. Antimicrobial monoglycerides for swine and poultry applications. *Front. Anim. Sci.* 3:1019320. <https://doi.org/10.3389/fanim.2022.1019320>.
- Kamande, G. M., J. Baah, K.-J. Cheng, T. A. McAllister, and J. A. Shelford. 2000. Effects of Tween 60 and Tween 80 on protease activity, thiol group reactivity, protein adsorption, and cellulose degradation by rumen microbial enzymes. *J. Dairy Sci.* 83:536–542. [https://doi.org/10.3168/jds.S0022-0302\(00\)74913-7](https://doi.org/10.3168/jds.S0022-0302(00)74913-7).
- Khiaosa-ard, R., M. Kreuzer, and F. Leiber. 2015. Apparent recovery of C18 polyunsaturated fatty acids from feed in cow milk: A meta-analysis of the importance of dietary fatty acids and feeding regimens in diets without fat supplementation. *J. Dairy Sci.* 98:6399–6414. <https://doi.org/10.3168/jds.2015-9459>.
- Kumar, A., S. K. Kheravii, L. Li, and S.-B. Wu. 2021. Monoglyceride blend reduces mortality, improves nutrient digestibility, and intestinal health in broilers subjected to clinical necrotic enteritis challenge. *Animals (Basel)* 11:1432. <https://doi.org/10.3390/ani11051432>.
- Lee, C., D. L. Morris, J. E. Copelin, J. M. Hettick, and I. H. Kwon. 2019. Effects of lysophospholipids on short-term production, nitrogen utilization, and rumen fermentation and bacterial population in lactating dairy cows. *J. Dairy Sci.* 102:3110–3120. <https://doi.org/10.3168/jds.2018-15777>.
- Lofsten, J. R., J. G. Linn, J. K. Drackley, T. C. Jenkins, C. G. Soderholm, and A. F. Kertz. 2014. *Invited review*: Palmitic and stearic acid metabolism in lactating dairy cows. *J. Dairy Sci.* 97:4661–4674. <https://doi.org/10.3168/jds.2014-7919>.
- Matamoros, C., R. N. Klopp, L. E. Moraes, and K. J. Harvatine. 2020. Meta-analysis of the relationship between milk trans-10 C18:1, milk fatty acids <16 C, and milk fat production. *J. Dairy Sci.* 103:10195–10206. <https://doi.org/10.3168/jds.2019-18129>.
- Pierce, R. B., R. Bomberger, and K. J. Harvatine. Unpublished. Examination of the effects of increasing oleic acid in a saturated fatty acid prill reacted with magnesium fed to high-producing Holstein cows.
- Porter, N. H., K. L. Clark, L. R. Rebelo, J. E. Copelin, I. H. Kwon, and C. Lee. 2024. Effects of saturated fatty acids with lysophospholipids on production and nutrient digestibility in lactating cows. *J. Dairy Sci.* 107:4524–4536. <https://doi.org/10.3168/jds.2023-24457>.
- Prom, C. M., J. M. dos Santos Neto, and A. L. Lock. 2022. Abomasal infusion of different exogenous emulsifiers alters fatty acid digestibility and milk fat yield of lactating dairy cows. *J. Dairy Sci.* 105:3102–3112. <https://doi.org/10.3168/jds.2021-21315>.
- Prom, C. M., J. M. dos Santos Neto, J. R. Newbold, and A. L. Lock. 2021. Abomasal infusion of oleic acid increases fatty acid digestibility and plasma insulin of lactating dairy cows. *J. Dairy Sci.* 104:12616–12627. <https://doi.org/10.3168/jds.2021-20954>.
- Rabiee, A. R., K. Breinhild, W. Scott, H. M. Golder, E. Block, and I. J. Lean. 2012. Effect of fat additions to diets of dairy cattle on milk production and components: A meta-analysis and meta-regression. *J. Dairy Sci.* 95:3225–3247. <https://doi.org/10.3168/jds.2011-4895>.
- Rico, D. E., and K. J. Harvatine. 2013. Induction of and recovery from milk fat depression occurs progressively in dairy cows switched between diets that differ in fiber and oil concentration. *J. Dairy Sci.* 96:6621–6630. <https://doi.org/10.3168/jds.2013-6820>.
- Rico, D. E., Y. Ying, and K. J. Harvatine. 2017. *Short communication*: Effects of lysolecithin on milk fat synthesis and milk fatty acid profile of cows fed diets differing in fiber and unsaturated fatty acid concentration. *J. Dairy Sci.* 100:9042–9047. <https://doi.org/10.3168/jds.2017-12976>.
- Sacakli, P., Ö. Ö. Çınar, A. Ceylan, M. S. Ramay, J. A. Harijaona, A. G. Bayraktaroglu, Y. Shastak, and A. Calik. 2023. Performance and gut health status of broilers fed diets supplemented with two graded levels of a monoglyceride blend. *Poult. Sci.* 102:102359. <https://doi.org/10.1016/j.psj.2022.102359>.
- Shepardson, R., and K. Harvatine. 2019. Effect of oleic acid and lecithin in saturated fatty acid supplements on production and nutrient digestibility in lactating dairy cows. *J. Dairy Sci.* 102(Suppl. 1):317. (Abstr.)
- Shpirer, J., L. Livshits, H. Kamer, T. Alon, Y. Portnik, and U. Moallem. 2023. The form more than the fatty acid profile of fat supplements influences digestibility but not necessarily the production performance of dairy cows. *J. Dairy Sci.* 106:2395–2407. <https://doi.org/10.3168/jds.2022-22190>.
- Sukhija, P. S., and D. L. Palmquist. 1988. Rapid method for determination of total fatty acid content and composition of feedstuffs and feces. *J. Agric. Food Chem.* 36:1202–1206. <https://doi.org/10.1021/jf00084a019>.
- Tomkins, T., and S. Lamb. 2014. Partial neutralization of free fatty acid mixtures with magnesium, livestock feed compositions including them, and methods of making same. XXX. US Pat. No. 8853435 B1.
- Van Soest, P. J., J. B. Robertson, and B. A. Lewis. 1991. Methods for dietary fiber, neutral detergent fiber, and nonstarch polysaccharides in relation to animal nutrition. *J. Dairy Sci.* 74:3583–3597. [https://doi.org/10.3168/jds.S0022-0302\(91\)78551-2](https://doi.org/10.3168/jds.S0022-0302(91)78551-2).
- Zhao, P. Y., Z. F. Zhang, R. X. Lan, W. C. Liu, and I. H. Kim. 2017. Effect of lysophospholipids in diets differing in fat contents on growth performance, nutrient digestibility, milk composition and litter performance of lactating sows. *Anim.* 11:984–990. <https://doi.org/10.1017/S1751731116002238>.

ORCID

- S. L. Burtnett, <https://orcid.org/0000-0001-8025-9044>
 Y. Adeniji, <https://orcid.org/0000-0002-2790-0362>
 P. Yoder, <https://orcid.org/0000-0001-6646-1991>
 M. F. Scott, <https://orcid.org/0009-0006-2329-8366>
 K. J. Harvatine <https://orcid.org/0000-0001-6422-2647>